Ishwar Soni

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Summary

Computer Vision and Machine Learning engineer with experience in human motion modeling, data pipelines, and large-scale datasets. Worked on SMPL/SMPL-H reconstruction and ML data cleaning systems, focusing on algorithm improvement, experimentation, and robust preprocessing pipelines. Strong in Python with working knowledge of data structures and algorithms.

Experience

Semantic Labs **Nov 2025 – Jan 2026**

Computer Vision / Motion Processing Intern *Remote*

- Analyzed and improved motion reconstruction algorithms in SMPL/SMPL-H pipelines, resolving orientation flips, scaling mismatches, and joint inconsistencies.
- Developed preprocessing pipelines for large-scale AMASS datasets with complex skeletal structures.
- Built motion cleaning pipelines reducing noise and improving temporal consistency across motion sequences.
- Conducted validation using visualization tools and debugging workflows for large-scale motion data.

Technical Skills

Languages: Python
Machine Learning: NumPy, Pandas, Scikit-learn, Model Evaluation
Computer Vision: Motion Analysis, Pose Estimation, SMPL, SMPL-H, BVH
Core CS: Data Structures, Algorithms, Problem Solving
Tools: Git, GitHub, FastAPI, Jupyter Notebook

Projects

BVH-to-SMPLH Motion Processing Pipeline *Python, NumPy, SciPy*

- Investigated BVH → SMPL-H conversion issues caused by coordinate misalignment and schema differences.
- Designed and improved transformation algorithms to resolve orientation flips and scaling errors.
- Improved motion stability and reduced jitter using smoothing, FK-based grounding, and foot-lock techniques.
- Built visualization tools to validate motion quality across large-scale datasets.

Datacleanr – ML Data Cleaning Framework *Python, Pandas*

- Designed a scalable data cleaning pipeline handling missing values, outliers, skewness, and feature redundancy.
- Implemented safety constraints to prevent data leakage and excessive data loss.
- Conducted experiments on 50+ datasets to evaluate robustness across diverse data distributions.
- Reduced manual data cleaning effort by automating repetitive preprocessing tasks.

VisionFX – Real-Time AR/VFX Engine *Python, OpenCV, MediaPipe, NumPy, ModernGL*

- Developed a real-time AR/VFX engine that overlays interactive visual effects on live camera feeds.
- Built gesture-controlled Rasengan and Chidori-inspired energy effects using hand tracking and custom rendering pipelines.
- Implemented MediaPipe-based hand landmark detection to accurately map visual effects to left and right hand movements.
- Designed a modular rendering and effects pipeline, enabling smooth integration and management of multiple real-time visual effects.

Certifications

Microsoft Applied Skills – Azure OpenAI
NVIDIA Certification – RAG Agent Development

Education

Techno India NJR Institute of Technology **2023 – 2027**

B.Tech in Computer Science *Udaipur, India*